

Math 199, Fall 2023
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Participation assignment 3 - Derivatives by definition

Estimated time: 45-60 minutes.

Point value: 3 points.

Goals: Understand how derivatives are defined and how to interpret what they mean in various situations. Learn how to use the definition to compute derivatives.

1) Let $f(x)$ be a function. Write a definition for the derivative $f'(x)$ of f at x as a limit.

2) For each of the following scenarios, describe what the derivative represents.

(a) $p(t)$ = the position of a person running at time t .

(b) $s(t)$ = the speed of a car merging onto the highway at time t .

(c) $f(x)$ = the y -coordinate of the curve $(x, f(x))$.

(d) $h(x)$ = the elevation of a hiking trail at the x -mile mark.

3) Use the definition of the derivative to compute the derivatives of the following functions.

(a) $f(x) = 2x^3 - 2x^2 + 1$ at $x = 1$.

(b) $g(t) = \frac{t-1}{t+1}$ at $t = -2$

(c) $\exp(x) = e^x$ for any x . (*Hint:* remember that $e = \lim_{h \rightarrow 0} (1+h)^{\frac{1}{h}}$.)

4) Identify each of the following limits as the derivative of a function $f(x)$ at a point $x = a$. Identify $f(x)$ and a explicitly.

(a) $\lim_{h \rightarrow 0} \frac{\sin(\frac{\pi}{2} + h) - 1}{h}$

(b) $\lim_{h \rightarrow 0} \frac{\ln(h^2 + 4h + 4) - \ln(4)}{h}$

(c) $\lim_{w \rightarrow 3} \frac{\sqrt{w} - \sqrt{3}}{w - 3}$

5) Find the slope of the line tangent to the graph of $f(x) = \frac{x}{x+1}$ at the origin.

6) There are various rules for computing derivatives of functions that are combinations of simpler functions. Derive the rules for sums and products using the definition of the derivative. i.e. Let $(f + g)(x) = f(x) + g(x)$ and $(f \cdot g)(x) = f(x)g(x)$.

(a) Derive a formula for $(f + g)'(x)$ in terms of $x, f(x), g(x), f'(x)$, or $g'(x)$.

(b) Derive a formula for $(f \cdot g)'(x)$ in terms of $x, f(x), g(x), f'(x)$, or $g'(x)$.